Regulation of Body Fluid Composition



Body Fluid Distribution

- The human body is around 60% water
- Distributed in different locations
- Continual exchange of fluid and materials between locations (compartments)

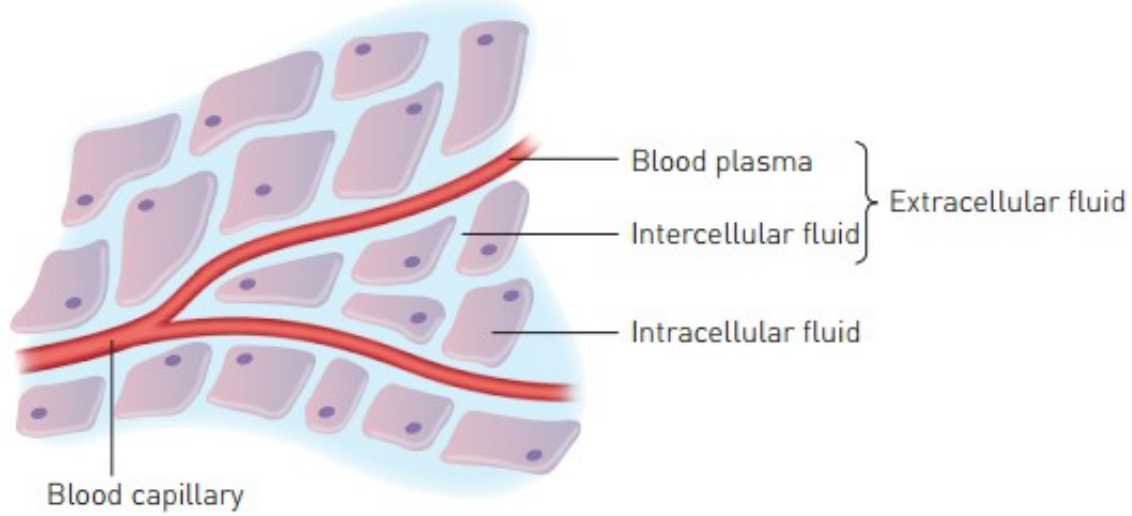
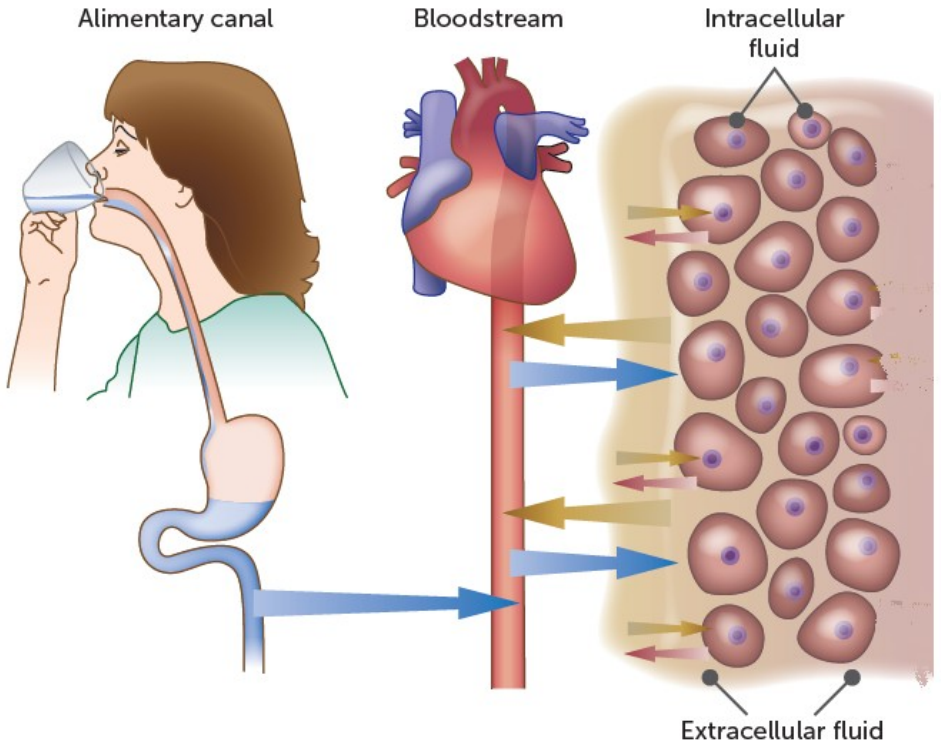


Figure 7.8 The body fluids

TYPE OF BODY FLUID	PROPORTION OF TOTAL BODY FLUID	COMPONENTS OF THE BODY FLUID
Intracellular fluid	2/3 of total body water	Fluid inside the cell – the cytosol
Extracellular fluid	1/3 of total body water	Fluid that is outside the cells
Plasma (intravascular fluid)	Approx. 1/4 of extracellular fluid	The fluid part of the blood
Interstitial fluid and transcellular fluid	Approx. 3/4 of extracellular fluid	Lymph, cerebrospinal fluid, synovial fluid in joints, fluids of eyes and ears, fluid in the chest and abdominal cavities and around the heart, fluids of the alimentary canal, kidney filtrate

Fluid Exchange

- Continuous exchange of fluids eg:
  - Environment to digestive system
  - Digestive system to blood
  - Between blood and intercellular (interstitial) fluid
  - Between interstitial fluid and lymph
  - Between interstitial fluid and cytoplasm
  - Between blood and excretory systems
  - Between excretory systems and environment



- Water and dissolved substances move freely through membranes
  - Diffusion
  - Osmosis

Fluid Balance

- Fluid gain must equal fluid loss
- Fluid composition must also remain steady
  - Ion concentration
  - Glucose concentration
  - Nitrogenous waste concentration
  - pH
  - Blood gas concentration

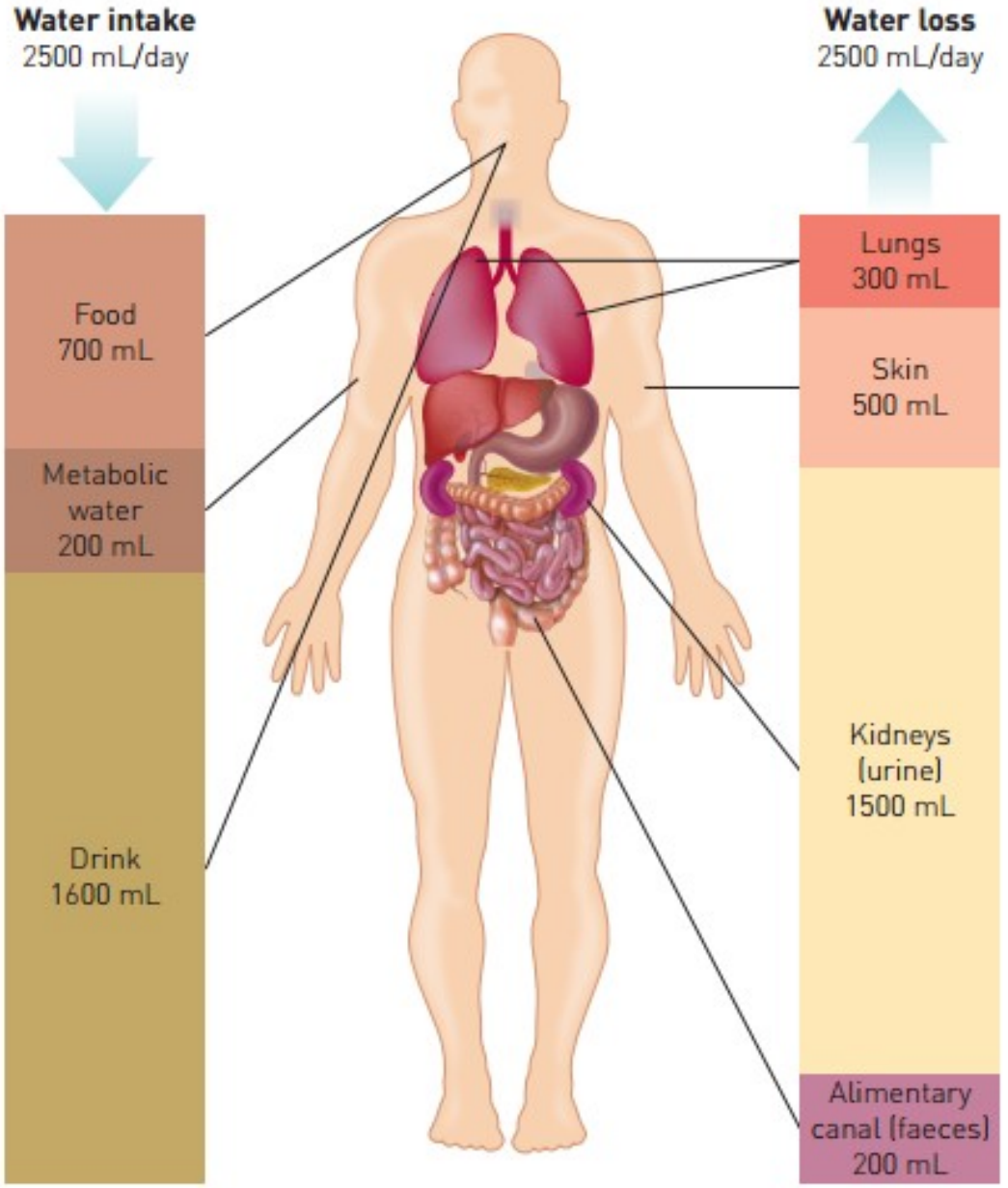


Figure 7.9 Daily fluid intake and output

Excretion and fluid balance

- Removal of metabolic products so they do not build up in body fluids
- Lungs:
  - Carbon dioxide excretion
  - Some water also lost by evaporation
- Sweat glands:
  - Secrete water, with salts, urea, lactic acid
- Digestive (alimentary) canal
  - Bile pigments, products of haemoglobin breakdown
  - Water also lost
- Kidneys
  - Main excretory organs
  - Maintain body fluid levels and balance solute concentrations
  - Urea removed also

Renal system

- Primary mechanism for regulation of body fluid composition.

- Filters blood, removes waste products and balances ion concentration.

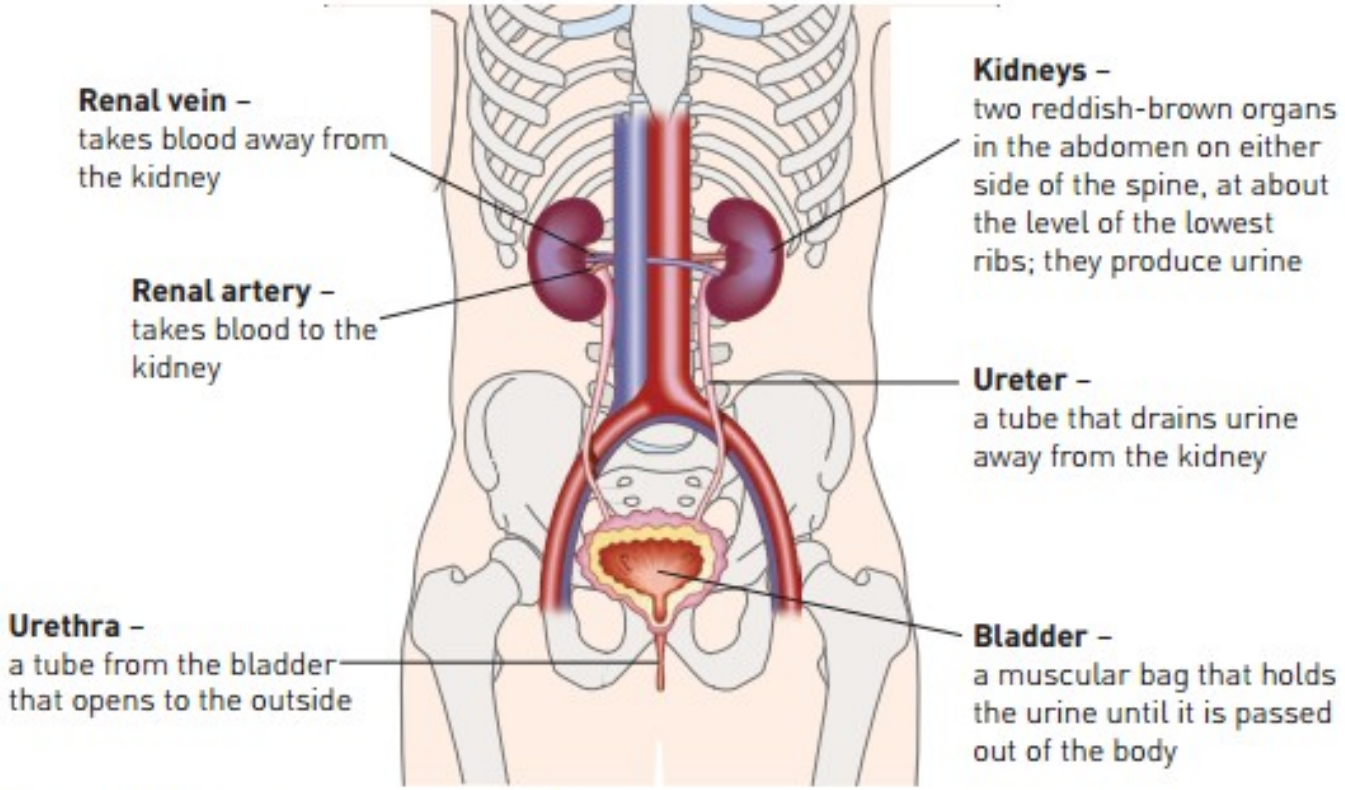


Figure 7.10 The kidneys and associated organs

The Nephron

- Functional Unit of the kidney.

FILTRATION

- Fluid component of blood filtered at glomerulus

REABSORPTION

- Water and useful substances are reabsorbed under homeostatic control to maintain constant body fluid composition

SECRETION

- Some additional substances are removed from blood into the kidney tubule.

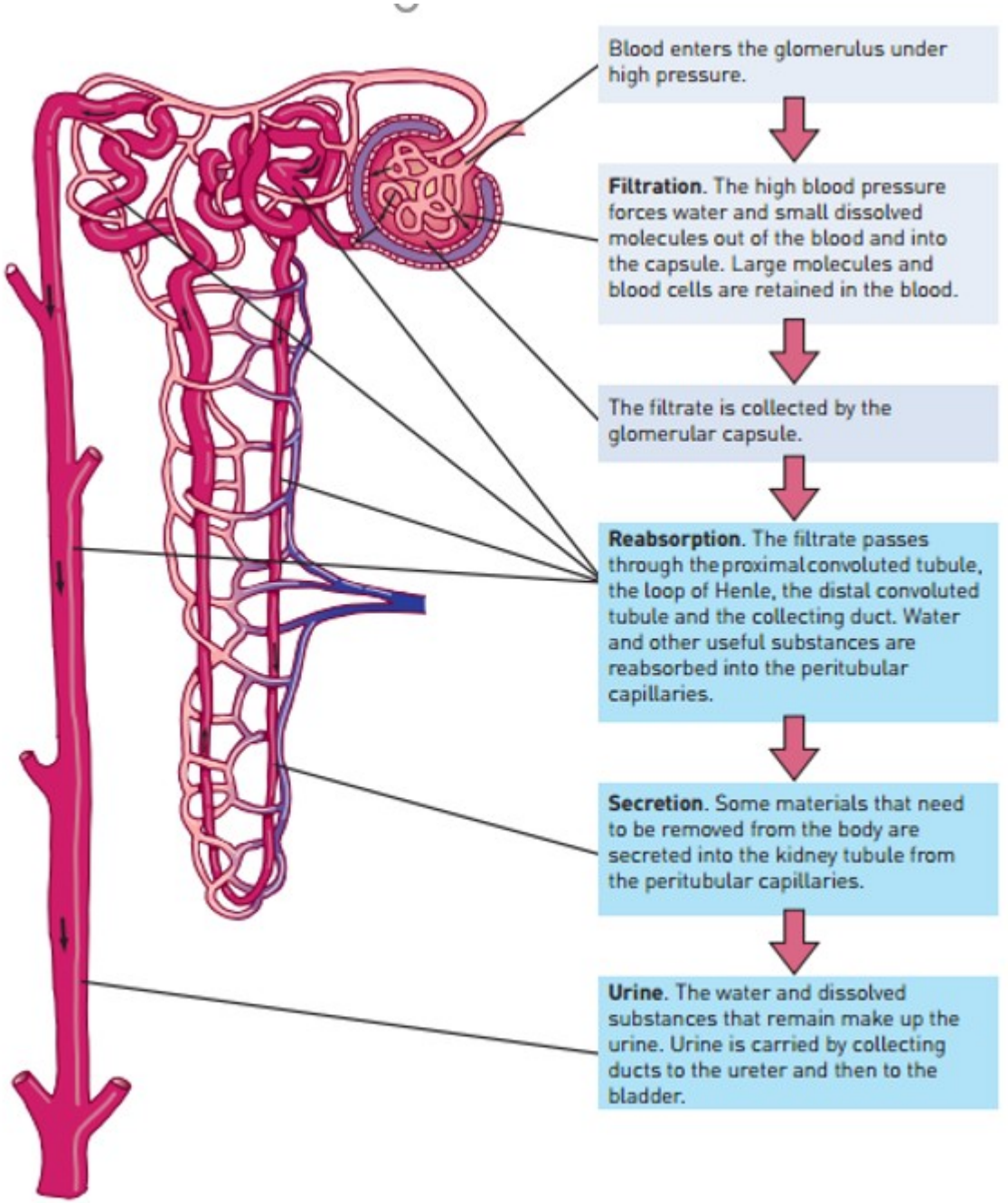


Figure 7.11 The functional unit of the kidney is the nephron.

Renal System Role in Body Fluid homeostasis

- Renal system balances secretion and reabsorption of water to maintain fluid volume:
  - Too little water: urine concentrated
  - Too much water: urine dilute, produced faster

- Reabsorption of water
  - Proximal convoluted tubule and loop of Henle: via osmosis
  - Distal convoluted tubule and collecting duct: active transport
  - Active reabsorption **controlled by ADH (Anti-diuretic Hormone)**

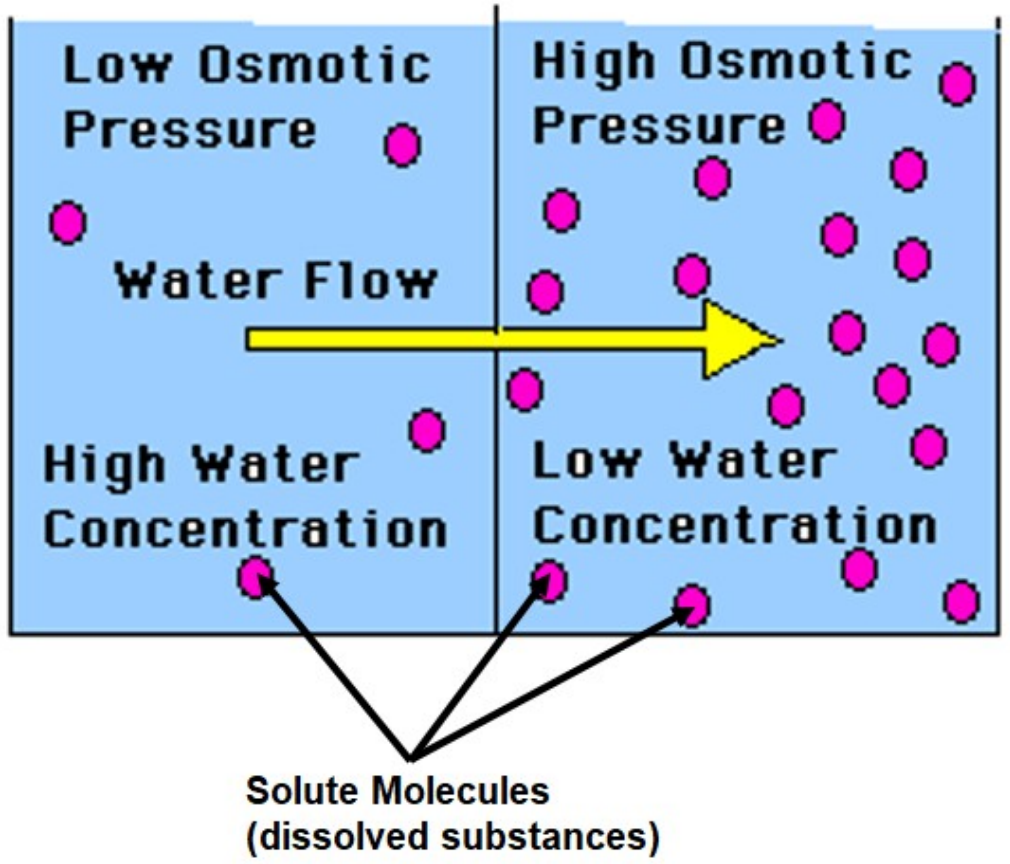
- Reabsorption of sodium ions
  - Distal convoluted tubule and collecting duct
  - **controlled by Aldosterone**
  - Increased Na+ reabsorption and K+ excretion via Na+/K+ pump
  - Water moves via osmosis with Na+

Osmotic Pressure

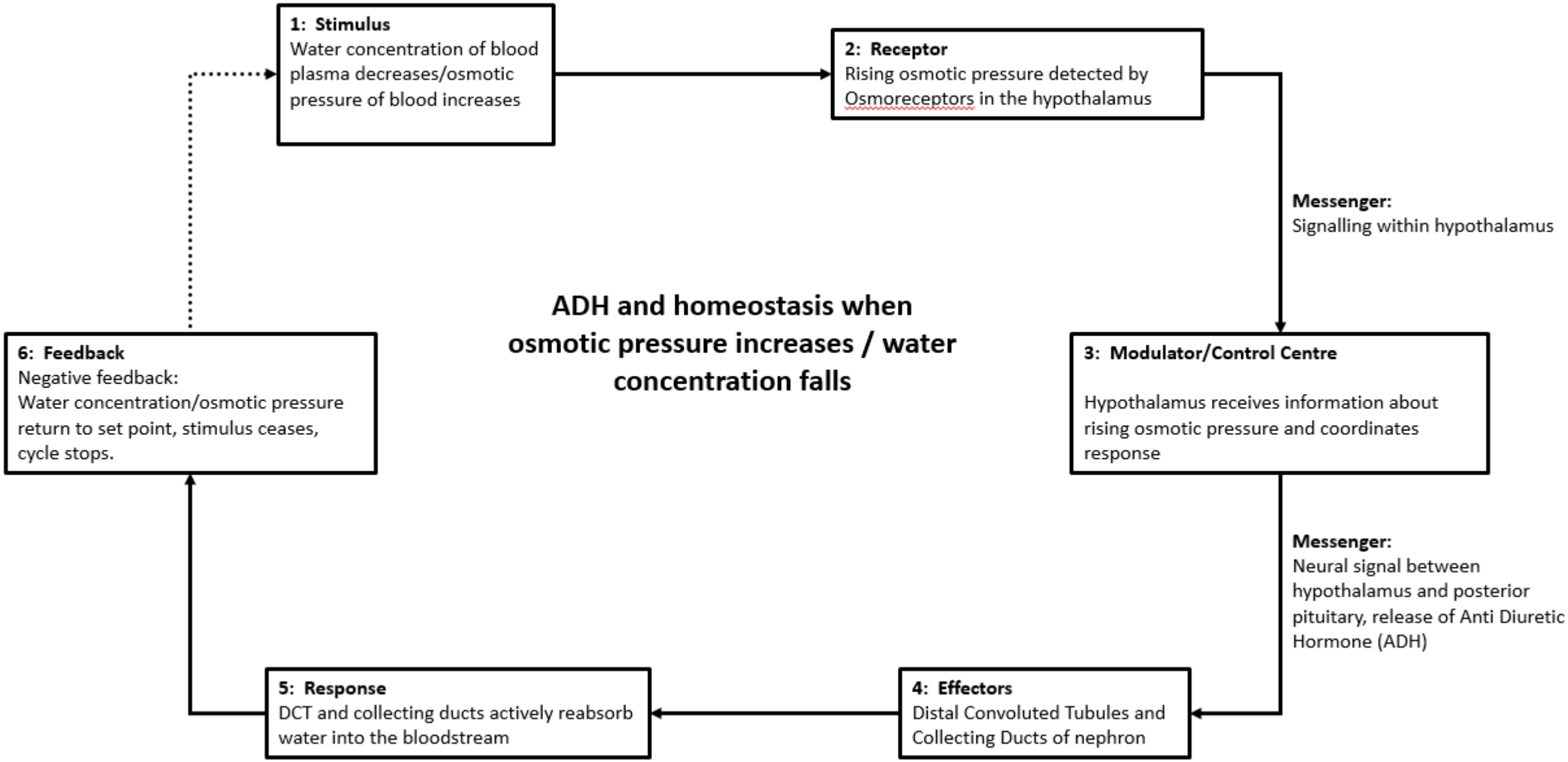
- Osmotic pressure is a measure of concentration of dissolved substances vs water.
- Body fluids with HIGH OSMOTIC PRESSURE have high solute concentration and low water concentration (like strong cordial)
- Body fluids with LOW OSMOTIC PRESSURE have low solute concentrations and high water concentration (like dilute cordial)

- *Osmoreceptors* in the **hypothalamus** detect osmotic pressure (ie how concentrated the solutes in the blood are)

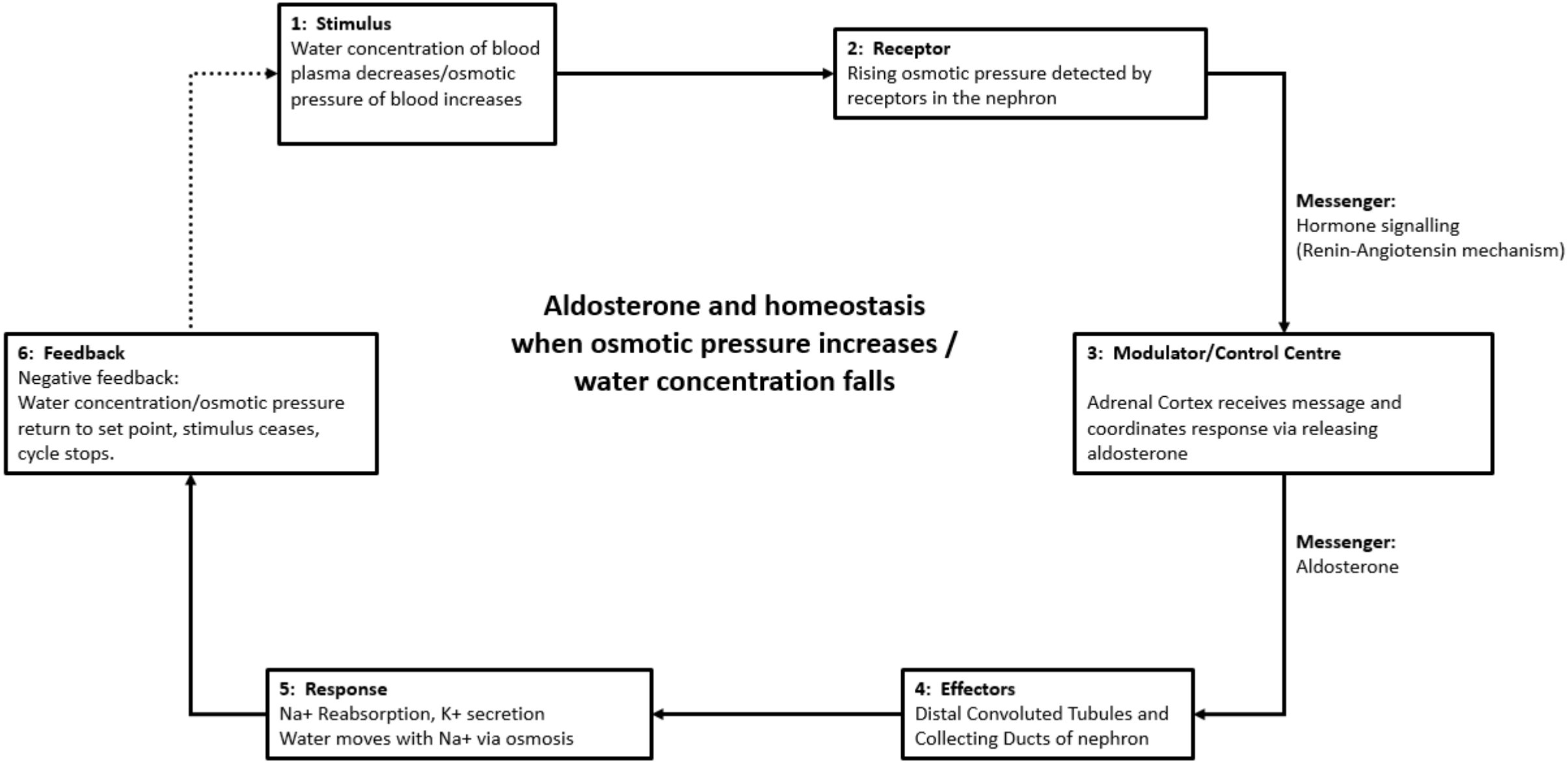
If the membrane between an area of low osmotic pressure and high osmotic pressure is semi-permeable, then water will flow from areas of high water concentration to areas of low water concentration. This is called osmosis.



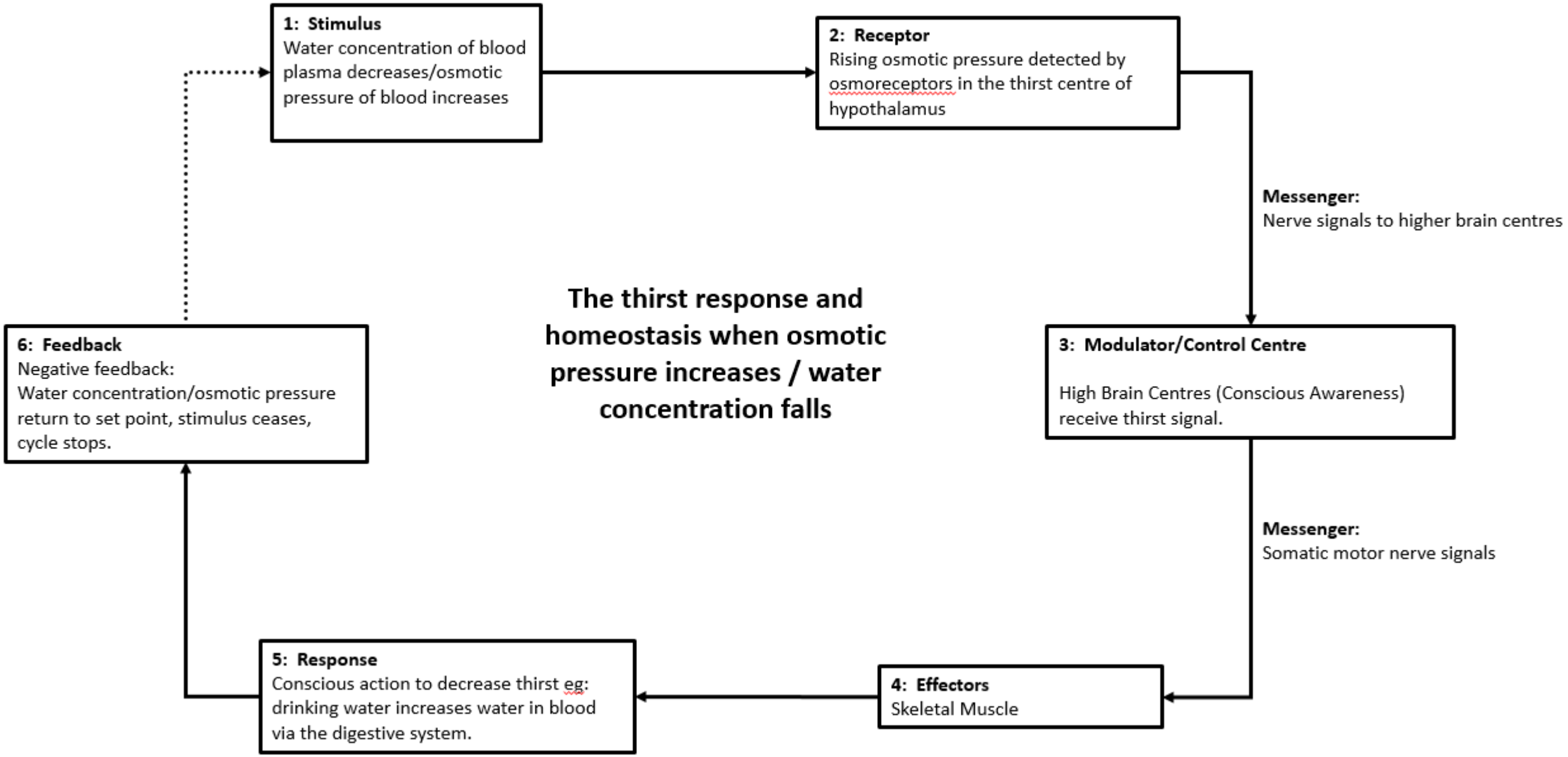
Homeostatic mechanisms when osmotic pressure increases: ADH



Homeostatic mechanisms when osmotic pressure increases: Aldosterone



Homeostatic mechanisms when osmotic pressure increases: Thirst Response



Homeostasis when fluid volume falls

Stimulus

Decreased blood volume and/or decreased Na+, increased osmotic pressure.



Receptor

Receptors in kidney tubules

Osmoreceptors in hypothalamus detect increased osmotic pressure

Message

Hormone release (renin-angiotensin mechanism)

Messaging within hypothalamus

Control Centre / Modulator

Adrenal Cortex

Hypothalamus/Posterior Pituitary

Thirst centre within hypothalamus

Message

aldosterone release

ADH release

Nerves within brain

Effector

Kidney tubules

Kidney tubules

Sweat glands

Skeletal Muscles

Response

Increased resorption of Na+ and water. Increased K+ and H+ secretion into urine

Increased permeability of kidney tubules (active reabsorption) therefore increased water reabsorption

Decreased sweat production

Increased thirst leads to increased drinking and fluid intake.

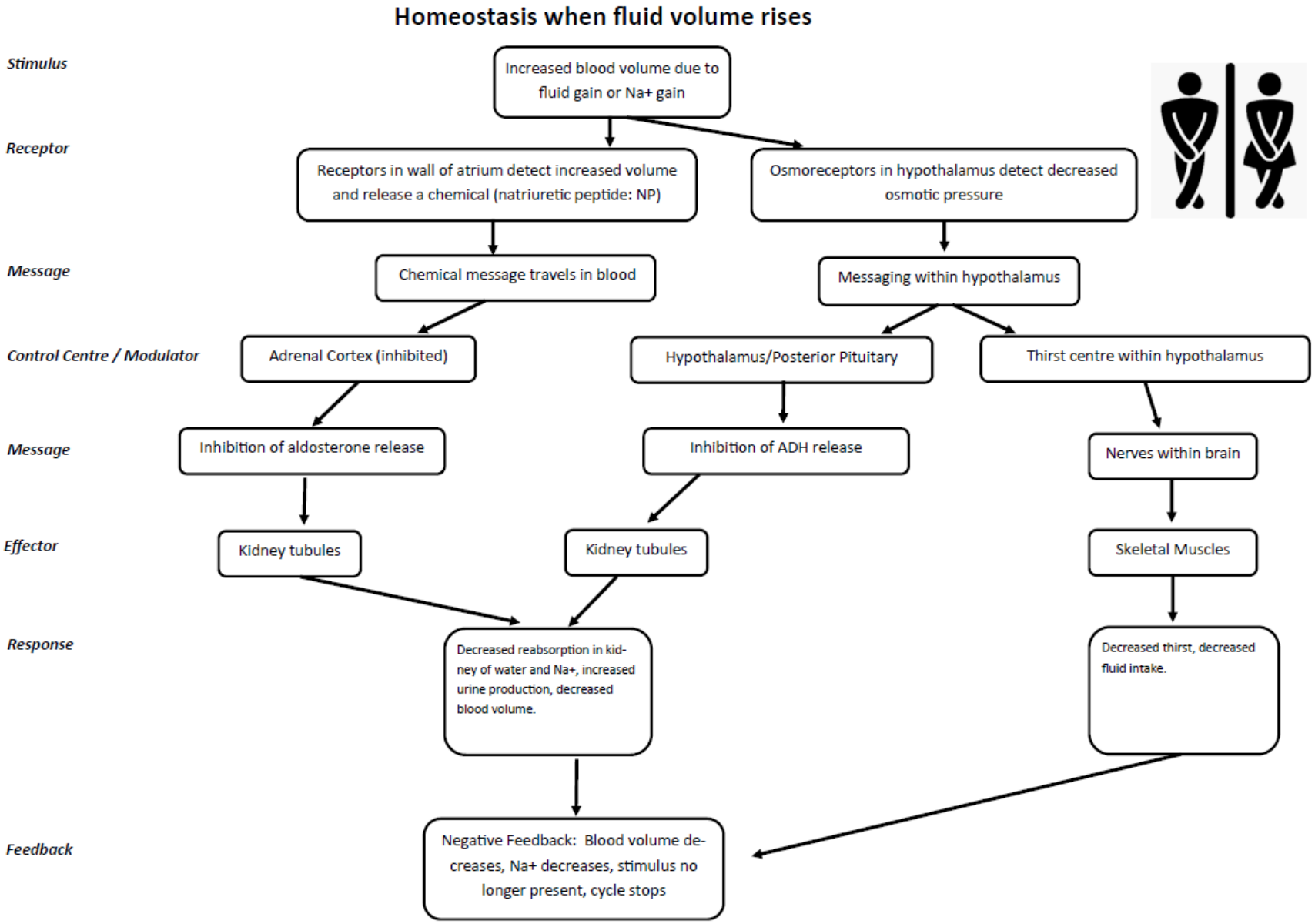
Feedback

Negative feedback: blood volume rises, stimulus no longer present, cycle stops

Homeostasis when water concentration rises/osmotic pressure falls.

- When fluid intake rises, water concentration rises, and osmotic pressure falls.

- Homeostatic response is to inhibit ADH and Aldosterone release, and the thirst response, so that water and Na+ are not reabsorbed, no new water is taken in, and the nephron can remove excess water into urine.



Increase in osmotic pressure beyond tolerance limits

- Dehydration
  - Noticeable when 2% of body water is lost
  - Due to:
    - Sweating
    - Vomiting
    - Diarrhoea
    - Evaporation from lungs etc
  - Symptoms
    - Severe thirst
    - Low blood pressure
    - Dizziness
    - Headache
    - Delirium
    - Unconsciousness
    - DEATH
  - Treatment
    - Replacement of fluids via IV drip with fluid AND electrolytes
    - Electrolyte drinks



Decrease in osmotic pressure beyond tolerance limits

- “Water Intoxication” (also called *hyponatremia)*
- Too much water intake
  - More than can be excreted by kidneys to maintain balance of fluid and salts
  - Occurs where water AND salts are lost, but only water is drunk to replace.
    - Marathon runners
    - Extreme athletes
  - Or where water is drunk well beyond thirst quenching needs
    - Idiots
  - Low ion concentration in blood and interstitial fluid causes water to move into cells via osmosis, causing cells to swell – this is a problem, especially in the brain!
  - Symptoms
    - Light-headedness
    - Headache, vomiting , collapse
    - DEATH
  - Treatment
    - Antidiuretic medication, but often fatal if severe.

